

STAAR-Aligned Science Lesson Plans — Grades 5 & 8

5 Fun & Engaging Lessons with STAAR Practice Built In

Lesson 1: Ecosystem Escape Room

Grade: 5th **TEKS:** 5.12A (Observe the way organisms live and survive in their ecosystem by interacting with the living and nonliving components), 5.12B (Describe how the flow of energy derived from the Sun, used by producers to create their own food, is transferred through a food chain and food web to consumers and decomposers) **Duration:** 60 minutes **Materials:** 5 "Lock Stations" (envelopes with puzzles), padlock props or combination sheets, ecosystem picture cards, food chain/food web diagrams, "Escape Room Passport" (printed), colored pencils, timer

Learning Objective

I can describe how energy flows through an ecosystem from producers to consumers to decomposers, and explain how organisms depend on living and nonliving components of their ecosystem.

STAAR Connection

STAAR Science frequently tests **food chains/webs, energy flow, and ecosystem relationships** — students must trace energy, predict what happens when a population changes, and identify producers/consumers/decomposers.

Warm-Up (8 minutes): "Ecosystem Brainstorm Race"

Teams of 4. Each team gets a large piece of paper divided into 4 quadrants:

- Quadrant 1: PRODUCERS (organisms that make their own food)
- Quadrant 2: CONSUMERS (organisms that eat other organisms)
- Quadrant 3: DECOMPOSERS (organisms that break down dead matter)
- Quadrant 4: NONLIVING THINGS in an ecosystem

2-minute timer: Teams brainstorm as many examples as possible in each quadrant.

Debrief: "Which quadrant was hardest? Why?" Most students struggle with decomposers and nonliving factors.

Key vocabulary review:

- **Producer:** Makes own food using sunlight (plants, algae)
 - **Consumer:** Eats other organisms (herbivore, carnivore, omnivore)
 - **Decomposer:** Breaks down dead organisms (fungi, bacteria, worms)
 - **Ecosystem:** All living and nonliving things in an area interacting together
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Main Activity (30 minutes): "The Great Ecosystem Escape"

Setup: 5 stations around the room. Teams must solve the puzzle at each station to get a "code" (answer). All 5 codes unlock the final challenge. Teams carry an "Escape Room Passport" to record answers.

Station 1: "Energy Flow Decoder" (Food Chain)

Puzzle: Arrange these organisms into a correct food chain by placing them in order of energy flow:

Cards: Hawk, Grasshopper, Sun, Grass, Frog

Correct order: **Sun → Grass → Grasshopper → Frog → Hawk**

Follow-up questions on the passport:

1. Which organism is the producer? (Grass)
2. Which is the primary consumer? (Grasshopper)
3. Where does ALL the energy in this food chain originally come from? (The Sun)
4. What would happen to the hawk population if a disease killed most of the frogs?

CODE = The number of consumers in this food chain (3)

Station 2: "Food Web Mystery" (Food Web Analysis)

Puzzle: Display a food web diagram showing:

- Grass → Rabbit → Fox
- Grass → Mouse → Snake → Hawk
- Grass → Grasshopper → Frog → Snake
- Dead organisms → Mushrooms (decomposer)

Questions:

1. Name TWO organisms that eat grass. (Rabbit, Mouse, Grasshopper)
2. If the mouse population disappeared, what would happen to the snake population? Why?

3. What role do the mushrooms play? Why are they important?
4. An animal that eats BOTH plants and animals is called an _____. Name one from this food web.

CODE = The number of arrows pointing TO the snake (2: mouse and frog)

Station 3: "Nonliving Factors Challenge"

Puzzle: Match each scenario to the nonliving factor that caused it:

Scenario	Nonliving Factor
Fish die in a pond during a long drought	A. Temperature
Plants grow taller on the south side of a building	B. Water
Lizards bask on rocks in the morning before hunting	C. Sunlight
Soil near a volcano is rich and supports lots of plant growth	D. Soil/minerals
Cacti have thick stems to store water in the desert	E. Water

Follow-up: "Give ONE example of how a nonliving factor in YOUR neighborhood affects a living organism."

CODE = Number of scenarios where WATER is the nonliving factor (2: drought + cacti)

Station 4: "Producer Power" (Photosynthesis Basics)

Puzzle: Fill in the blanks in the photosynthesis equation using word cards:

_____ + _____ + _____ → _____ + _____

Word cards: Sunlight, Water, Carbon Dioxide, Glucose (food), Oxygen

Correct: **Sunlight + Water + Carbon Dioxide → Glucose + Oxygen**

Questions:

1. Why are producers called the "base" of every food chain?
2. If pollution blocked sunlight from reaching ocean algae, what would happen to the ENTIRE ocean food web? Explain step by step.
3. Producers use the Sun's energy to make _____ (glucose/food), which consumers get by _____ (eating the producers or eating animals that ate producers).

CODE = Number of INPUTS (things needed) for photosynthesis (3: sunlight, water, CO2)

Station 5: "Ecosystem Disruption" (Critical Thinking)

Puzzle — Scenario Analysis:

A new highway was built through a forest, splitting it in two. Scientists noticed the following changes over the next 5 years:

- Deer could no longer cross to find food on the other side
- The deer population on one side decreased
- The wolf population decreased because there were fewer deer
- Small plants grew more because fewer deer were eating them
- Rabbit population increased because of more plants
- The fox population increased because of more rabbits

Questions:

1. Draw a simple diagram showing how the highway created a chain reaction through the ecosystem.
2. Was the highway a living or nonliving change to the ecosystem? (Nonliving)
3. **STAAR Constructed Response:** Explain how a change to ONE population in an ecosystem can affect other populations. Use the highway scenario as an example, and include at least TWO cause-and-effect relationships from the text.

CODE = Total number of populations mentioned in the scenario (5: deer, wolves, plants, rabbits, foxes)

FINAL CHALLENGE (after all 5 codes):

Codes: 3, 2, 2, 3, 5

Use the codes: Add all codes together ($3+2+2+3+5 = 15$). The team that correctly gets 15 and can explain ALL five station concepts first escapes!

Practice (12 minutes): STAAR-Format Questions

Question 1 (Multi-Select):

Which of the following are TRUE about energy flow in an ecosystem?
Select THREE.

- ☐ Energy flows from consumers to producers
- ☐ All energy in an ecosystem originally comes from the Sun
- ☐ Producers convert sunlight into food through photosynthesis
- ☐ Decomposers add energy to the ecosystem
- ☐ Energy is transferred from one organism to another through food chains
- ☐ Consumers make their own food

Correct: Sun is original source ✓; Producers convert sunlight ✓; Energy transferred through food chains ✓

Question 2 (Constructed Response):

In an ocean ecosystem, phytoplankton (tiny plants) are eaten by small fish, which are eaten by larger fish, which are eaten by sharks. If pollution caused the phytoplankton population to decrease dramatically, predict what would happen to the shark population. Explain your reasoning step by step.

Question 3:

A student observes that plants near a stream grow taller than plants far from the stream. What nonliving factor BEST explains this observation?
A) Temperature B) Water availability C) Soil type D) Sunlight

Answer: B

Exit Ticket (5 minutes)

Name the THREE roles organisms play in an ecosystem. Give one example of each. Then explain in one sentence why decomposers are essential.

Differentiation

Group

Modification

ELL

Provide illustrated vocabulary cards (producer, consumer, decomposer with pictures). Station instructions available in Spanish. Pair with bilingual partner. Food chain diagrams use pictures, not just words.

SpEd

Reduce to 3 stations (1, 2, 5). Provide word bank for all fill-in-the-blank. Pre-label food chain cards with roles (producer, consumer, etc.). Allow verbal responses recorded by partner. Provide a food chain reference sheet.

GT

At Station 5, research and present a real-world example of ecosystem disruption (invasive species, deforestation). Calculate: "If only 10% of energy passes from one level to the next, and the Sun provides 10,000 units to producers, how much reaches a tertiary consumer?" Explore biomagnification.

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Lesson 2: Kitchen Chemistry

Grade: 8th TEKS: 8.5E (Investigate how evidence of chemical reactions indicates that new substances with different properties are formed), 8.5F (Recognize whether a chemical equation containing coefficients is balanced or not and how that relates to

the law of conservation of mass) **Duration:** 60 minutes **Materials:** Baking soda, vinegar, clear cups/beakers, balloons, ziplock bags, steel wool, water, hydrogen peroxide (3%), dry yeast, dish soap, food coloring, graduated cylinders, "Lab Report" sheets, safety goggles, paper towels

Learning Objective

I can identify evidence of chemical reactions and explain how the law of conservation of mass applies to chemical changes.

STAAR Connection

STAAR Science tests students on **identifying evidence of chemical reactions** (gas production, color change, temperature change, precipitate formation) and understanding **conservation of mass**.

Warm-Up (8 minutes): "Chemical or Physical?"

Display 6 scenarios. Students sort them into CHEMICAL CHANGE or PHYSICAL CHANGE on dry erase boards:

1. Burning a piece of paper → **Chemical** (new substances: ash, CO₂, smoke)
2. Melting an ice cube → **Physical** (still water, just liquid)
3. Rusting a nail → **Chemical** (iron + oxygen = iron oxide — new substance)
4. Cutting a piece of wood → **Physical** (still wood, just smaller)
5. Baking a cake → **Chemical** (raw ingredients become something new — can't unbake it!)
6. Dissolving sugar in water → **Physical** (sugar is still there — evaporate water and it comes back)

Key question: "How do you KNOW something is a chemical change? What EVIDENCE would you look for?"

Post the evidence list:

Evidence of Chemical Reactions: 🔥 Temperature change (gets hot or cold) 💨 Gas production (bubbles, fizzing, odor) 🌈 Color change
Precipitate forms (solid appears in liquid) ⚡ Light or sound produced
🚫 Cannot be easily reversed

Main Activity (30 minutes): "Kitchen Chemistry Lab Rotation"

Safety review (2 min): Goggles on. No tasting. Report spills immediately. Wash hands after.

4 Lab Stations — Teams of 4-5, rotating every 7 minutes.

Station 1: "Balloon Blow-Up" (Baking Soda + Vinegar)

Procedure:

1. Pour 50 mL of vinegar into a clear cup
2. Put 2 spoonfuls of baking soda into a balloon (use a funnel)
3. Stretch the balloon over the cup opening **WITHOUT** letting baking soda fall in
4. Lift the balloon to dump the baking soda into the vinegar
5. Observe!

Lab Report Questions:

1. What did you observe? (Fizzing, bubbling, balloon inflated)
 2. What evidence of a chemical reaction did you see? (Gas production — CO₂ inflated the balloon)
 3. The chemical equation is: $\text{NaHCO}_3 + \text{CH}_3\text{COOH} \rightarrow \text{NaCH}_3\text{COO} + \text{H}_2\text{O} + \text{CO}_2$
 - What gas inflated the balloon? (Carbon dioxide — CO₂)
 4. If you weighed the cup+vinegar+baking soda **BEFORE** and **AFTER**, would the total weight change? Why or why not? → **If sealed (balloon captures gas): weight stays the same — conservation of mass! → If open: weight decreases because CO₂ gas escapes into the air**
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Station 2: "Elephant Toothpaste" (Hydrogen Peroxide + Yeast)

Procedure:

1. Pour 100 mL of hydrogen peroxide (3%) into a tall cup
2. Add a few drops of food coloring and a squirt of dish soap
3. In a separate cup, mix 1 tablespoon of dry yeast with 3 tablespoons of warm water. Stir for 30 seconds.
4. Pour the yeast mixture into the hydrogen peroxide cup
5. Stand back and observe!

Lab Report Questions:

1. What happened? (Foam erupted out of the cup, got warm)
 2. List **ALL** evidence of a chemical reaction you observed. (Temperature increase, gas production/foam, rapid change)
 3. The yeast acts as a **catalyst** — it speeds up the reaction without being used up. The hydrogen peroxide breaks down into water and oxygen gas: $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$
 4. Why did the foam feel warm? (The reaction released energy — it's exothermic)
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Station 3: "Rusty Steel Wool" (Oxidation)

Procedure:

1. Observe a piece of new steel wool — record its color, texture, and flexibility
2. Observe a piece of steel wool that was soaked in vinegar and left out for 2 days (pre-prepared by teacher)
3. Compare the two pieces

Lab Report Questions:

1. Describe the differences between the new and old steel wool. (Color change: silver to reddish-brown; texture: rougher; flexibility: more brittle)
 2. What evidence of a chemical reaction do you see? (Color change, new substance formed)
 3. The chemical equation: $4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$
 - o Count the iron (Fe) atoms on each side. Are they balanced? (Left: 4 Fe; Right: $2 \times 2 = 4$ Fe ✓)
 - o Count the oxygen (O) atoms. Are they balanced? (Left: $3 \times 2 = 6$ O; Right: $2 \times 3 = 6$ O ✓)
 4. This reaction normally takes weeks. Why did soaking in vinegar speed it up? (Vinegar removed the protective coating on the steel wool, exposing more iron to oxygen)
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Station 4: "Conservation of Mass Challenge"**Procedure:**

1. Place a ziplock bag on a scale. Record the mass: _____ g
2. Put 50 mL of vinegar in the bag. Place a small cup with baking soda inside the bag (don't mix yet!)
3. Seal the bag completely. Record the mass: _____ g
4. Tip the cup over inside the sealed bag to mix the vinegar and baking soda
5. Observe the reaction. After it finishes, record the mass: _____ g

Lab Report Questions:

1. Mass before reaction: _____ g
 2. Mass after reaction: _____ g
 3. Did the mass change? (It should be the same or very close!)
 4. **EXPLAIN:** Why does the mass stay the same even though a gas was produced? (The bag is sealed — the gas is still inside. Atoms are not created or destroyed in a chemical reaction; they just rearrange. This is the **law of conservation of mass**.)
 5. **Predict:** What would happen to the measured mass if you OPENED the bag after the reaction? Why? (Mass would decrease because CO_2 gas would escape)
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Practice (12 minutes): STAAR-Format Questions**Question 1 (Multi-Select):**

A student mixes two clear liquids together. Which of the following observations would indicate a chemical reaction occurred? Select ALL that apply.

- ☐ The mixture turns yellow
- ☐ The mixture gets warm
- ☐ The liquids are fully mixed together
- ☐ Bubbles form in the mixture
- ☐ The total volume increases
- ☐ A solid forms at the bottom of the container

Correct: Turns yellow (color change) ✓; Gets warm (temp change) ✓; Bubbles (gas production) ✓; Solid forms (precipitate) ✓

Question 2 (Constructed Response):

A student places 50 grams of a substance in an open container and burns it. After the burning is complete, only 35 grams of ash remain.

The student concludes that mass was destroyed during the reaction. Is the student correct? Explain why or why not, using the law of conservation of mass in your answer.

The student is incorrect. Mass was NOT destroyed — it was converted to gases (CO₂, water vapor, smoke) that escaped into the air. If the reaction happened in a sealed container, the total mass would still be 50 grams. The law of conservation of mass states that matter cannot be created or destroyed in a chemical reaction.

Question 3:

Consider the equation: $_\text{Fe} + _\text{O}_2 \rightarrow _\text{Fe}_2\text{O}_3$ Which set of coefficients correctly balances this equation? A) 1, 1, 1 B) 2, 3, 1 C) 4, 3, 2 D) 3, 2, 1

Answer: C — $4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$ (4 Fe each side, 6 O each side)

Exit Ticket (5 minutes)

You mix baking soda and vinegar in a sealed container on a scale. The scale reads 200g before the reaction.

Part A: What will the scale read after the reaction? Explain. Part B: Name TWO pieces of evidence you would observe that show a chemical reaction occurred.

Differentiation

Group	Modification
ELL	Lab procedure cards with step-by-step pictures. Vocabulary cards: chemical reaction, conservation of mass, reactant, product — with Spanish translations. Allow lab reports partially in native language.
SpEd	Pair with a strong partner. Provide pre-filled lab reports (students complete observation sections only). Focus on Stations 1 and 4 (most concrete). Provide evidence checklist to circle rather than recall. Simplify balancing equations to counting atoms only.
GT	Calculate the actual moles of CO ₂ produced in Station 1. Research and explain catalysts in more depth (Station 2). Write balanced equations for additional reactions. Design their own experiment to demonstrate conservation of mass. Explore endothermic vs. exothermic reactions.

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Lesson 3: Weather Forecasters

Grade: 5th **TEKS:** 5.8A (Differentiate between weather and climate), 5.8B (Explain how the Sun and the ocean interact in the water cycle) **Duration:** 60 minutes

Materials: Weather data charts (printed), blank weather maps of Texas (printed), colored pencils, "Meteorologist Report" sheets, thermometers (optional), water cycle diagram pieces, spray bottles (optional demo), chart paper

Learning Objective

I can explain the difference between weather and climate, describe the water cycle, and use data to make weather predictions.

STAAR Connection

STAAR tests the **difference between weather and climate** and understanding of the **water cycle** — students must apply these concepts to scenarios, not just define them.

Warm-Up (8 minutes): "Weather vs. Climate — Quick Sort"

Students sort statements into WEATHER or CLIMATE on dry erase boards:

1. "It's 95°F and sunny today in Austin." → **Weather**
2. "Houston gets about 50 inches of rain per year." → **Climate**
3. "A thunderstorm is expected this afternoon." → **Weather**
4. "The Sahara Desert is hot and dry." → **Climate**
5. "It might snow in Dallas next Tuesday." → **Weather**
6. "Alaska has long, cold winters." → **Climate**

Discuss: "What's the RULE for telling them apart?"

Anchor chart:

Weather = what's happening in the atmosphere RIGHT NOW or in the near future (hours/days) **Climate** = the PATTERN of weather in an area over a LONG TIME (years/decades) Memory trick: **Climate is what you EXPECT; weather is what you GET.**

Main Activity (25 minutes): "Meteorologist Training"

Phase 1: The Water Cycle (10 min)

Interactive build: Students receive cut-out labels and arrows to build the water cycle on a large diagram:

Components to place:

- **Evaporation:** Sun heats water in oceans/lakes → water turns to water vapor (gas)
- **Condensation:** Water vapor cools in the atmosphere → forms clouds (tiny water droplets)
- **Precipitation:** Water droplets in clouds get heavy → fall as rain, snow, sleet, or hail
- **Collection/Runoff:** Water flows into rivers, lakes, oceans → cycle starts again
- **Transpiration:** Plants release water vapor through their leaves

Key questions:

1. "What provides the ENERGY to drive the water cycle?" (The Sun — it heats water to evaporate)
2. "How does the OCEAN play a role?" (Most evaporation happens from oceans — they're the biggest water source)
3. "Is the water cycle creating NEW water?" (No! The same water has been cycling for billions of years)

Demo (optional): Spray warm water into a cold glass container. Students observe condensation forming. "That's a mini-cloud!"

Phase 2: Weather Forecaster Challenge (15 min)

Teams receive a "Weather Data Package" for a fictional Texas city:

5-Day Weather Data:

Day	High Temp (°F)	Low Temp (°F)	Humidity	Wind Direction	Cloud Cover	Barometric Pressure
Mon	78	62	45%	South	Clear	30.2 (high)

Tue	82	65	55%	South	Partly Cloudy	30.0
Wed	75	60	75%	North	Cloudy	29.7 (dropping)
Thu	68	55	90%	North	Overcast	29.3 (low)
Fri	?	?	?	?	?	?

Meteorologist Report:

1. Describe the weather TREND you see from Monday to Thursday. (Getting cooler, more humid, cloudier, pressure dropping)
2. What do you think the weather will be like on Friday? Why? (Likely rain — high humidity, low pressure, overcast skies, cold front moving through from the north)
3. A cold front is moving through. What evidence in the data supports this? (Wind shifted from South to North, temperature dropped, pressure dropped)
4. Draw weather symbols on a blank Texas map showing Friday's predicted weather.

STAAR Constructed Response:

Using the data above, explain how you can predict that it will likely rain on Friday. Include at least THREE pieces of evidence from the data table.

Practice (17 minutes): STAAR-Format Questions

Question 1 (Multi-Select):

Which of the following are examples of CLIMATE (not weather)? Select ALL.

- ☐ It rained 2 inches yesterday
- ☐ The average July temperature in San Antonio is 95°F
- ☐ A tornado watch was issued for Dallas County
- ☐ The Texas coast receives 40-60 inches of rain annually
- ☐ Seattle is known for rainy, mild weather
- ☐ Tomorrow's high will be 72°F

Correct: Average July temp ✓; Annual rainfall ✓; Seattle known for rainy weather ✓

Question 2 (Constructed Response):

Explain how the Sun and the ocean work together in the water cycle. Describe at least TWO steps of the water cycle in your answer, and explain the Sun's role in each step.

Question 3:

A student notices that after a rainy week, the puddle in the school parking lot slowly disappears even though it doesn't rain again. What process of the water cycle explains this? A) Condensation B) Precipitation C) Evaporation D) Collection

Answer: C

Question 4:

How is weather different from climate? Give an example of each for your city.

Exit Ticket (5 minutes)

Part A: Put these water cycle steps in order: Precipitation, Evaporation, Condensation, Collection Part B: Which step would be MOST affected if the Sun suddenly became less intense? Explain why.

Order: Evaporation → Condensation → Precipitation → Collection; Evaporation would be most affected because the Sun's energy drives evaporation.

Differentiation

Group

Modification

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|-------------|--|
| ELL | Water cycle vocabulary cards with pictures in English/Spanish. Provide labeled water cycle diagram for reference. Weather data table with picture icons for cloud cover. Sentence stems for meteorologist report. |
| SpEd | Pre-labeled water cycle diagram (students draw arrows only). Reduce weather data to 3 days. Provide multiple choice for Friday's prediction instead of open-ended. Use physical water cycle model (bag on window). |
| GT | Research and present: "How does climate change affect the water cycle?" Analyze real weather data from weather.gov for their city. Explore the relationship between air pressure and weather patterns in depth. Compare weather patterns in different climate zones. |
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Lesson 4: Force & Motion Olympics

Grade: 8th **TEKS:** 8.6A (Demonstrate and calculate how unbalanced forces change the speed or direction of an object's motion), 8.6B (Differentiate between speed, velocity, and acceleration) **Duration:** 60 minutes **Materials:** Toy cars, ramps (textbooks + cardboard), rulers/measuring tape, stopwatches, marbles, spring scales

(optional), "Olympic Record Sheets" (printed), calculators, masking tape, textbooks for ramps

Learning Objective

I can differentiate between speed, velocity, and acceleration, and demonstrate how unbalanced forces affect motion.

STAAR Connection

STAAR tests **calculating speed** (d/t), understanding **velocity** (speed + direction), and predicting effects of **unbalanced forces** on motion.

Warm-Up (8 minutes): "Speed, Velocity, or Acceleration?"

Definitions review (quick — students should have prior knowledge):

- **Speed** = how fast something moves (distance ÷ time; no direction)
- **Velocity** = speed WITH direction (50 mph NORTH)
- **Acceleration** = change in velocity over time (speeding up, slowing down, OR changing direction)

Sort these scenarios:

1. "A car travels 60 miles in 1 hour." → **Speed** (60 mph)
2. "A car travels 60 mph heading east." → **Velocity** (speed + direction)
3. "A car goes from 0 to 60 mph in 10 seconds." → **Acceleration** (velocity is changing)
4. "A car makes a turn at a constant 30 mph." → **Acceleration** (direction changed!)
5. "A runner completes a 100m dash in 12 seconds." → **Speed** ($100 \div 12 \approx 8.3$ m/s)

Key point #5 is tricky! "Wait — the car's speed didn't change, but it's STILL acceleration? YES! Because acceleration is any change in velocity, and velocity includes direction."

Main Activity (30 minutes): "Force & Motion Olympics"

5 Olympic Events — teams rotate through stations (6 min each).

Event 1: "The Speed Trial" (Calculate Speed)

Procedure:

1. Set up a straight track on the floor (use masking tape) — 3 meters long
2. Roll a toy car down the track

3. Use a stopwatch to time how long it takes to travel 3 meters
4. Calculate speed: **Speed = Distance ÷ Time**
5. Do 3 trials. Calculate average speed.

Record Sheet:

Trial	Distance (m)	Time (s)	Speed (m/s)
1	3.0		
2	3.0		
3	3.0		
Average			

Questions:

1. What was your car's average speed?
2. If your car traveled at this speed for 30 seconds, how far would it go? (Speed × 30)
3. Is this measurement speed or velocity? Why? (Speed — we didn't specify direction)
4. How would you turn this into velocity? (Add direction: "0.75 m/s toward the east wall")

Event 2: "The Ramp Race" (Unbalanced Forces)

Procedure:

1. Build a ramp using textbooks and cardboard (start at 2-book height)
2. Release a toy car from the top. Time how long it takes to reach the bottom + travel 1 meter past the ramp.
3. Repeat with 4-book height
4. Repeat with 6-book height

Record Sheet:

Ramp Height	Time to Travel Course (s)	Speed (m/s)	Faster or Slower?
2 books			—
4 books			
6 books			

Questions:

1. What happened to the car's speed as the ramp got taller? (Increased)
2. What FORCE caused the car to speed up? (Gravity — pulling it down the ramp)
3. Was the force on the car balanced or unbalanced? How do you know? (Unbalanced — the car's motion changed/accelerated)

4. **Predict:** What would happen with a 10-book ramp? (Even faster — greater gravitational force component along the ramp)
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Event 3: "The Collision Course" (Force & Direction Change)

Procedure:

1. Roll a marble straight across a table at moderate speed
2. Roll a second marble so it hits the first one from the side
3. Observe what happens to BOTH marbles' direction
4. Repeat 3 times

Questions:

1. What happened to Marble 1's direction after being hit? (Changed direction)
 2. What happened to Marble 2? (Also changed direction and/or slowed)
 3. Did the force cause a change in speed, direction, or both? (Both!)
 4. **Constructed Response:** Explain how this demonstrates that an unbalanced force changes an object's velocity. Use the words "force," "velocity," and "direction" in your answer.
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Event 4: "The Friction Challenge" (Unbalanced Forces)

Procedure:

1. Roll a toy car across a smooth desk surface. Measure distance traveled.
2. Roll the SAME car with the SAME push across carpet/rough surface. Measure distance.
3. Roll the SAME car across a towel. Measure distance.

Record Sheet:

Surface	Distance Traveled (cm)
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Smooth desk

Carpet

Towel

Questions:

1. On which surface did the car travel farthest? Why?
 2. What force caused the car to slow down and stop? (Friction)
 3. Was friction a balanced or unbalanced force? (Unbalanced — it changed the car's speed)
 4. If there were NO friction at all, what would happen to the car? (It would never stop! → Newton's 1st Law connection)
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Event 5: "The Acceleration Calculator"

Problem-solving station (no lab — calculate!):

1. A skateboarder goes from 0 m/s to 8 m/s in 4 seconds. What is the acceleration?

◦ $a = (\text{final velocity} - \text{initial velocity}) \div \text{time} = (8-0) \div 4 = 2 \text{ m/s}^2$

2. A car slows from 20 m/s to 5 m/s in 3 seconds. What is the acceleration?

◦ $a = (5-20) \div 3 = -5 \text{ m/s}^2$ (negative = deceleration!)

3. A ball rolling at 3 m/s accelerates at 2 m/s² for 5 seconds. What is the final speed?

◦ $v = 3 + (2 \times 5) = 13 \text{ m/s}$

4. **Constructed Response:** A student says "my car is accelerating because it's going fast." Is the student's reasoning correct? Explain the difference between speed and acceleration.

Practice (12 minutes): STAAR-Format Questions

Question 1:

A runner completes a 400-meter race in 50 seconds. What is the runner's average speed? A) 4 m/s B) 8 m/s C) 20 m/s D) 80 m/s

Answer: B ($400 \div 50 = 8$)

Question 2 (Multi-Select):

Which of the following are examples of acceleration? Select ALL.

- ☐ A car traveling at a constant 65 mph on a straight highway
- ☐ A ball being thrown upward and slowing down
- ☐ A car making a U-turn at constant speed
- ☐ A rocket launching from the ground
- ☐ A book sitting still on a desk
- ☐ A bicycle rider speeding up going downhill

Correct: Ball slowing ✓; U-turn (direction change) ✓; Rocket launching ✓; Bicycle speeding up ✓

Question 3 (Constructed Response):

Two students push a box. Student A pushes with 20 N to the right. Student B pushes with 15 N to the left. Will the box move? If so, in which direction and with what net force? Explain using the concept of unbalanced forces.

Net force = 20 - 15 = 5 N to the right. The forces are unbalanced, so the box will accelerate to the right.

Exit Ticket (5 minutes)

Part A: Calculate the speed of a car that travels 150 meters in 10 seconds.

Part B: The car then turns a corner at the same speed. Is the car accelerating during the turn? Explain why or why not.

Part A: 15 m/s; Part B: Yes — direction changed, which means velocity changed, which means acceleration occurred.

Differentiation

Group

Modification

- ELL** Vocabulary cards with visuals: speed (speedometer), velocity (arrow with number), acceleration (car speeding up). Formula reference card. Allow lab reports partially in native language.
- SpEd** Provide formula cards (Speed = d/t , Acceleration = $\Delta v/t$). Pre-fill lab report structure. Reduce to Events 1, 2, and 4. Allow calculator use. Pair with strong partner for calculations. Provide worked examples for Event 5.
- GT** Calculate acceleration from ramp data (Event 2). Explore Newton's Second Law ($F=ma$). Design an experiment to test: "Does mass affect acceleration down a ramp?" Graph speed vs. ramp height — what pattern emerges?
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Lesson 5: Space Mission Planner

Grade: 5th **TEKS:** 5.9A (Describe the physical properties, locations, and movements of the Sun, planets, moons, meteors, asteroids, and comets), 5.9B (Demonstrate that Earth rotates on its axis once approximately every 24 hours causing the day/night cycle and the apparent movement of the Sun across the sky) **Duration:** 60 minutes
Materials: Solar system fact cards (provided below), "Mission Planner" sheets (printed), rulers, colored pencils, globe, flashlight, styrofoam balls (optional for models), calculator, poster paper

Learning Objective

I can describe the physical properties and movements of objects in our solar system and explain how Earth's rotation causes day and night.

STAAR Connection

STAAR tests **solar system knowledge** (planet order, properties) and **Earth's rotation/revolution** — students must explain day/night and seasons, not just recall facts.

Warm-Up (8 minutes): "Planet Lineup"

Speed challenge: Students have 2 minutes to write all 8 planets in order from the Sun on their dry erase board.

Mercury - Venus - Earth - Mars - Jupiter - Saturn - Uranus - Neptune

Memory trick: **My Very Excellent Mother Just Served Us Nachos**

Then: "One BONUS fact about each planet — ready?"

Quick fire (teacher reads, students write T/F):

1. Mercury is the hottest planet. → **FALSE** (Venus is — greenhouse effect!)
 2. Mars is called the Red Planet. → **TRUE** (iron oxide/rust on surface)
 3. Jupiter is the largest planet. → **TRUE**
 4. Saturn is the only planet with rings. → **FALSE** (Jupiter, Uranus, Neptune have them too — Saturn's are just the most visible)
 5. Earth is the only planet with liquid water on its surface. → **TRUE** (that we know of!)
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Main Activity (25 minutes): "Mission to the Solar System"

Phase 1: Planet Research (10 min)

Each team of 4 is assigned a "space mission" to one planet (or 2 planets). They receive a Planet Fact Card and must complete a "Mission Report."

Planet Fact Cards (abbreviated — full cards would include images):

Mercury: Closest to Sun. 88-day year. No atmosphere. Extreme temps: 800°F day, -290°F night. No moons. Rocky planet.

Venus: 2nd from Sun. 225-day year. Thick CO₂ atmosphere (greenhouse effect — hottest planet at 900°F!). Rotates backwards. No moons. Rocky.

Mars: 4th from Sun. 687-day year. Thin atmosphere. -80°F average. 2 moons (Phobos, Deimos). Rocky. Has the largest volcano (Olympus Mons).

Jupiter: 5th from Sun. 12-year orbit. Gas giant. Great Red Spot (giant storm). 95 known moons (including Europa — possible ocean under ice!). Largest planet.

Saturn: 6th from Sun. 29-year orbit. Gas giant. Famous rings (ice and rock). 146 known moons (including Titan — has atmosphere). Second largest.

Uranus: 7th from Sun. 84-year orbit. Ice giant. Tilted on its side (98°). 27 moons. Blue-green color (methane).

Neptune: 8th from Sun. 165-year orbit. Ice giant. Strongest winds (1,200+ mph). 16 known moons. Blue color (methane).

Mission Report:

MISSION: Explore _____ (planet name)

1. LOCATION: ____ planet from the Sun

2. PHYSICAL PROPERTIES:

- Type (rocky/gas/ice giant): _____
- Approximate temperature: _____
- Atmosphere: _____
- Number of moons: _____
- One unique feature: _____

3. TRAVEL PLAN:

- How long is one year on this planet? _____
- Compare to Earth: that's _____ Earth years

4. MISSION CHALLENGE:

Could humans survive on this planet? Why or why not?
(Consider: temperature, atmosphere, gravity, water)

5. STAAR QUESTION – Write a multiple choice question about your planet for another team to answer:

Phase 2: Day & Night Demonstration (8 min)

Materials: Globe + flashlight (or lamp)

Teacher demonstration:

1. Darken the room. Shine flashlight on globe.
2. "What do you notice?" (One side is lit — that's day. The other side is dark — that's night.)
3. Slowly spin the globe. "Watch Texas. What happens?" (It goes from day to night to day again.)
4. "How long does one full spin take?" (About 24 hours — that's one day!)
5. "Does the Sun move across our sky, or do WE move?" (WE rotate — the Sun appears to move, but it's actually Earth spinning)

Key concepts:

- Earth **rotates** on its axis → causes **day and night** (24 hours)

- Earth **revolves** around the Sun → causes **seasons and year** (365 days)
- The Sun **APPEARS** to move across the sky (rises east, sets west) because Earth rotates

Student activity: On a diagram of Earth, draw and label:

- The axis (tilted line through poles)
- Arrow showing rotation direction (counterclockwise when viewed from North Pole)
- Where it's day (side facing Sun)
- Where it's night (side facing away)
- Where it's sunrise and sunset

Phase 3: Mission Presentations (7 min)

Each team gives a 60-second "Mission Briefing" to the class about their planet. Class takes notes on a solar system chart.

Practice (17 minutes): STAAR-Format Questions

Question 1 (Multi-Select):

Which of the following correctly describe physical properties of objects in our solar system? Select ALL that apply.

- ☐ Venus is the hottest planet because it is closest to the Sun
- ☐ Jupiter is a gas giant and the largest planet
- ☐ Earth rotates on its axis once every 24 hours
- ☐ Mars has liquid water on its surface
- ☐ Saturn has rings made of ice and rock
- ☐ Comets orbit the Sun and develop tails when they get close to it

Correct: Jupiter largest/gas giant ✓; Earth rotates every 24 hrs ✓; Saturn rings ✓; Comets develop tails ✓ (Venus is NOT hottest because it's closest — it's 2nd from Sun, hottest due to greenhouse effect)

Question 2 (Constructed Response):

A student in Texas notices that the Sun appears to move across the sky from east to west during the day. Explain what is **ACTUALLY** causing this apparent movement. Use the word "rotation" in your answer.

*The Sun is not actually moving across the sky — Earth is rotating on its axis from west to east. This rotation makes the Sun **APPEAR** to move from east to west. Earth completes one full rotation approximately every 24 hours, which is why we experience day and night.*

Question 3:

Earth orbits the Sun once every 365 days. Mars orbits the Sun once every 687 days. What is the BEST explanation for this difference? A) Mars rotates more slowly B) Mars is farther from the Sun and has a longer orbital path C) Mars is smaller than Earth D) Mars has a thinner atmosphere

Answer: B

Question 4:

A comet, an asteroid, and a meteor enter our solar system region. Match each to its description:

- Object made of ice and dust that develops a tail near the Sun: ____ (Comet)
- Rocky object that orbits the Sun, mostly found between Mars and Jupiter: ____ (Asteroid)
- A streak of light caused by a small space rock burning up in Earth's atmosphere: ____ (Meteor)

Exit Ticket (5 minutes)

Part A: Earth spins on its axis once every _____ hours. This causes _____ and _____. Part B: Explain why it's daytime in Texas but nighttime in China at the same moment.

Part A: 24 hours; day and night. Part B: Earth is a sphere that rotates. When Texas faces the Sun (day), the opposite side of Earth (where China is) faces away from the Sun (night).

Differentiation

Group

Modification

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| ELL | Planet fact cards in English and Spanish. Solar system vocabulary cards with images. Allow mission report in native language with key terms in English. Use visual models (globe + flashlight) extensively. |
| SpEd | Assign inner rocky planets only (simpler properties). Provide sentence frames for Mission Report. Use a pre-made solar system diagram for reference. Day/night diagram partially completed — students add labels. Partner with strong reader for fact cards. |
| GT | Research and compare dwarf planets (Pluto, Ceres, Eris). Calculate scale distances between planets. Explain WHY Venus is hotter than Mercury despite being farther from the Sun. Create a scale model of the solar system with correct relative sizes. Explore current Mars rover missions. |